COMPUTER ARCHITECTURE

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Basic Input/output

Lecture Outline

- External Devices
- I/O Modules
- Programmed I/O
- An Example of a RISC-Style I/O Program
- An Example of a CISC-Style I/O Program

I/O System Design Tool

- In addition to the processor and a set of memory modules
  - Set of I/O modules
- Each module is an interface to the system bus or
  - Central switch and controls one or more peripheral devices.
- I/O module is not a simply a set of mechanical connectors
  - Rather, it contains logic for performing a communication function between the peripheral and the bus.

I/O System Design Tool

- Why one does not connect peripherals directly to the system bus
  - Wide variety of peripherals with various methods of operation.
  - Data transfer rate of peripherals is often much slower and some of them faster.
  - Peripherals often use different data formats and word lengths.
- Need I/O modules

I/O System Design Tool

- This module has two functions
  - Interface to the processor and memory via the system bus or central switch.
  - Interface to one or more peripheral devices by tailored data links.

Cont.

Load R2, DATAIN

- reads the data from the DATAIN register and loads them into processor register R2. Similarly, the instruction

Store R2, DATAOUT

- sends the contents of register R2 to location DATAOUT, which is a register in an output device.

Device interface

- An I/O device is connected to the interconnection network by using a circuit, called the *device interface*
- which provides the means for data transfer and for the exchange of status and control information needed to facilitate the data transfers and govern the operation of the device.
- The interface includes some registers that can be accessed by the processor

Program-controlled I/O

- Consider a task that reads characters typed on a keyboard, stores these data in the memory, and displays the same characters on a display screen.
- A simple way of implementing this task is to write a program that performs all functions needed to realize the desired action. This method is known as *program-controlled I/O*.

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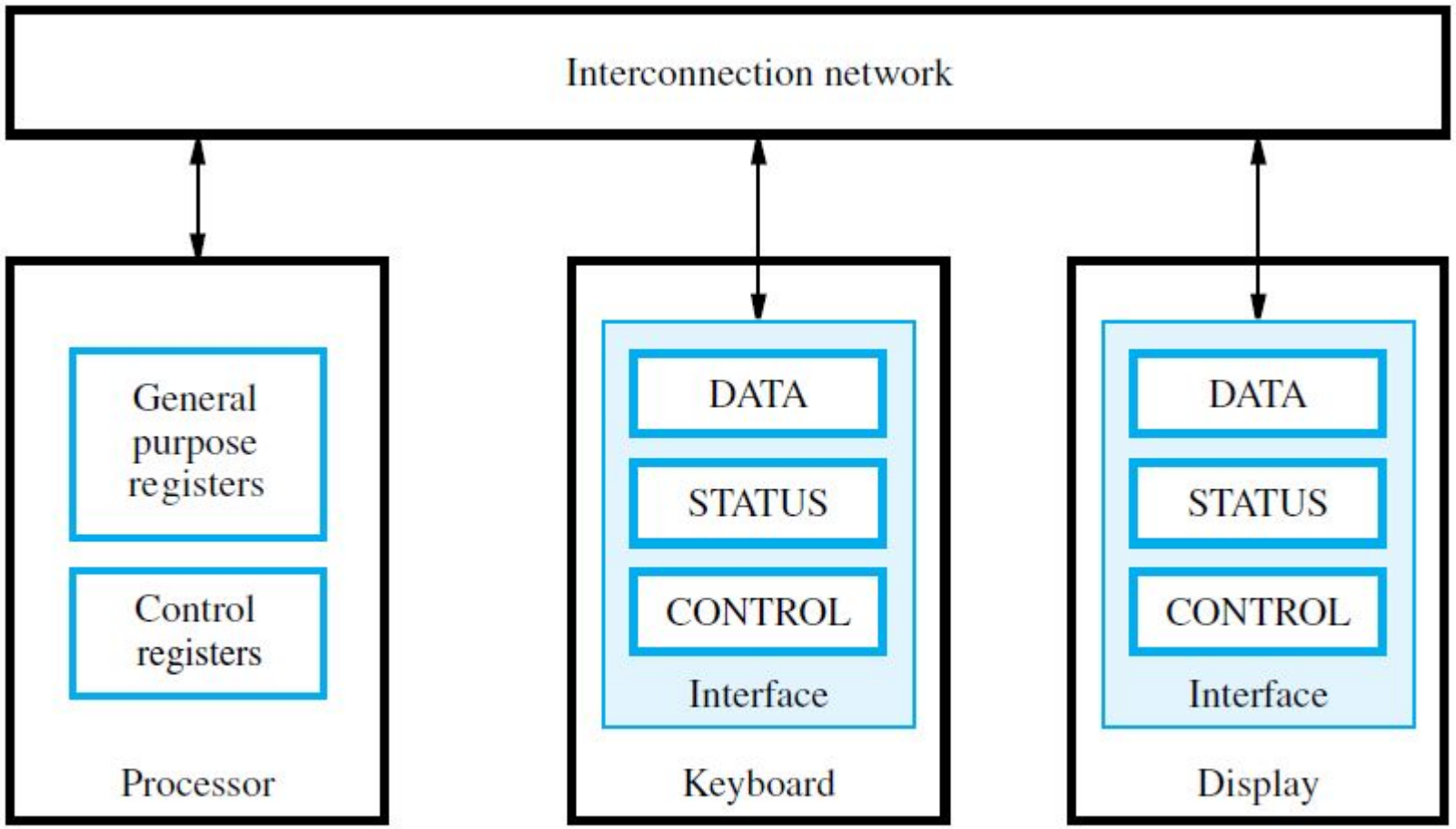


Figure 3.2 The connection for processor, keyboard, and display.

Cont..

- The rate of data transfer from the keyboard to a computer is limited by the typing speed of the user, which is unlikely to exceed a few characters per second.
- The rate of output transfers from the computer to the display is much higher.
- The difference in speed between the processor and I/O devices creates the need for mechanisms to synchronize the transfer of data between them.

Cont..

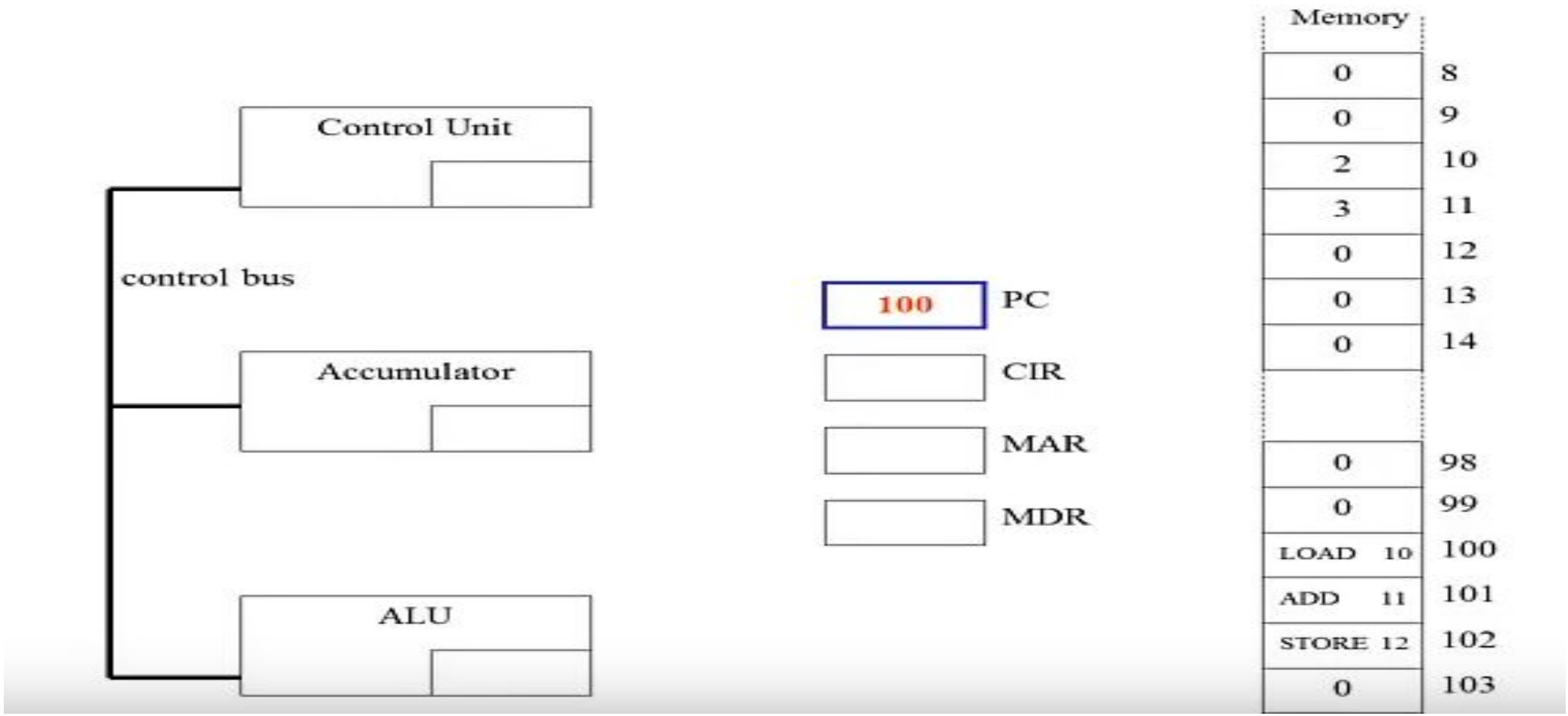
- For example, the contents of the keyboard character buffer KBD_DATA can be transferred to register R5 in the processor by the instruction

  LoadByte R5, KBD_DATA

- Similarly, the contents of register R5 can be transferred to DISP_DATA by the instruction

  StoreByte R5, DISP_DATA

What happen when the program runs



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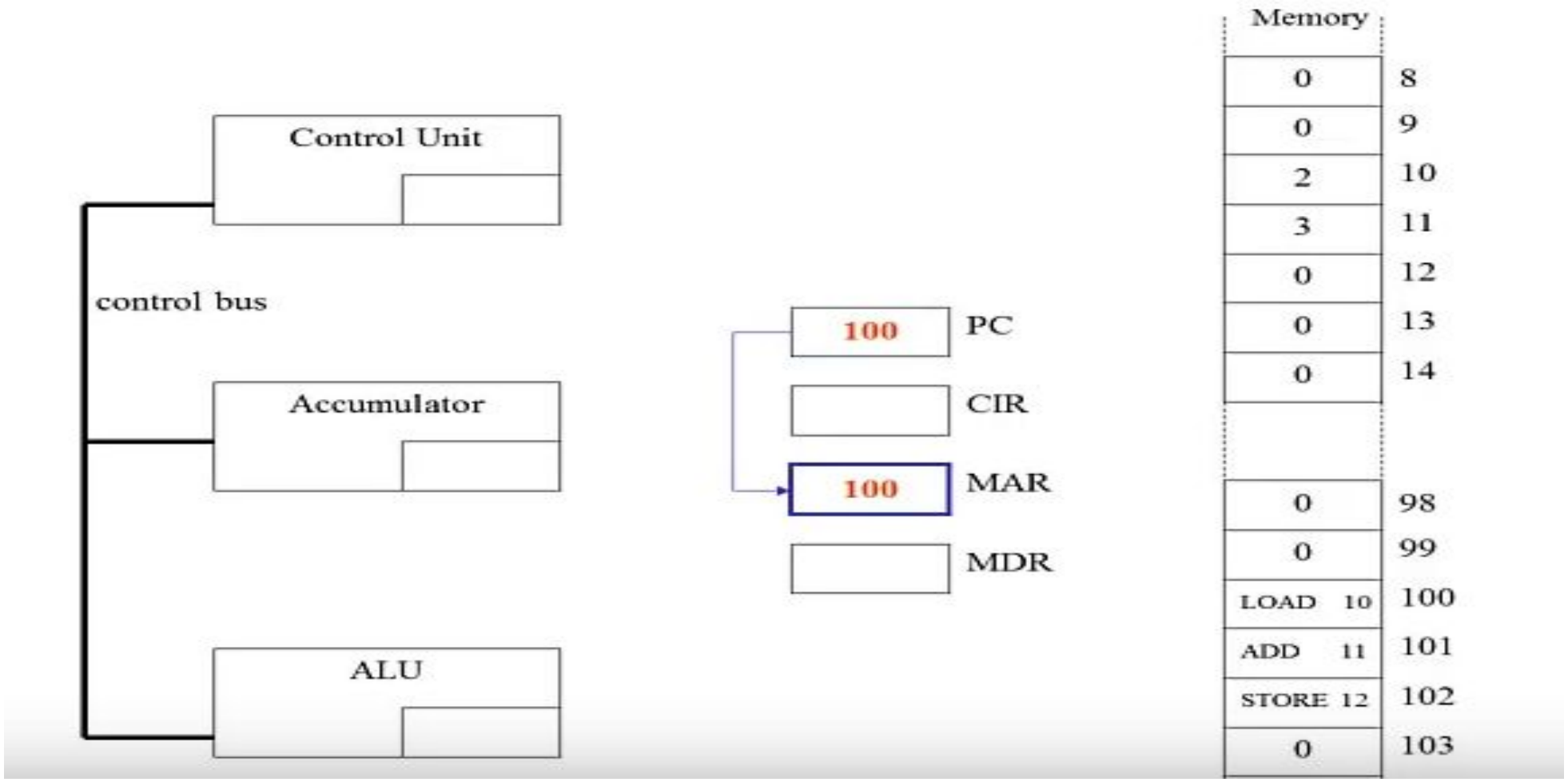

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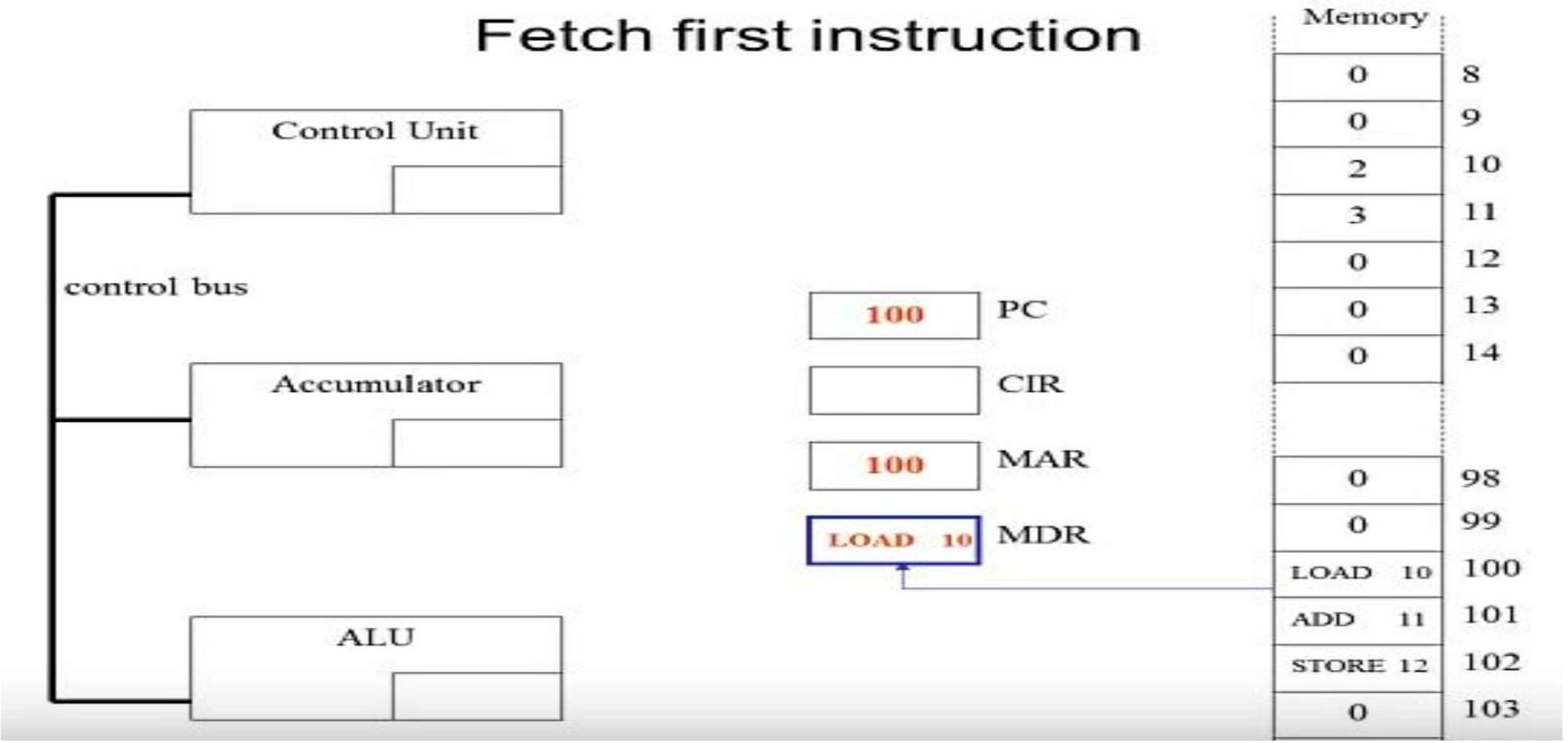

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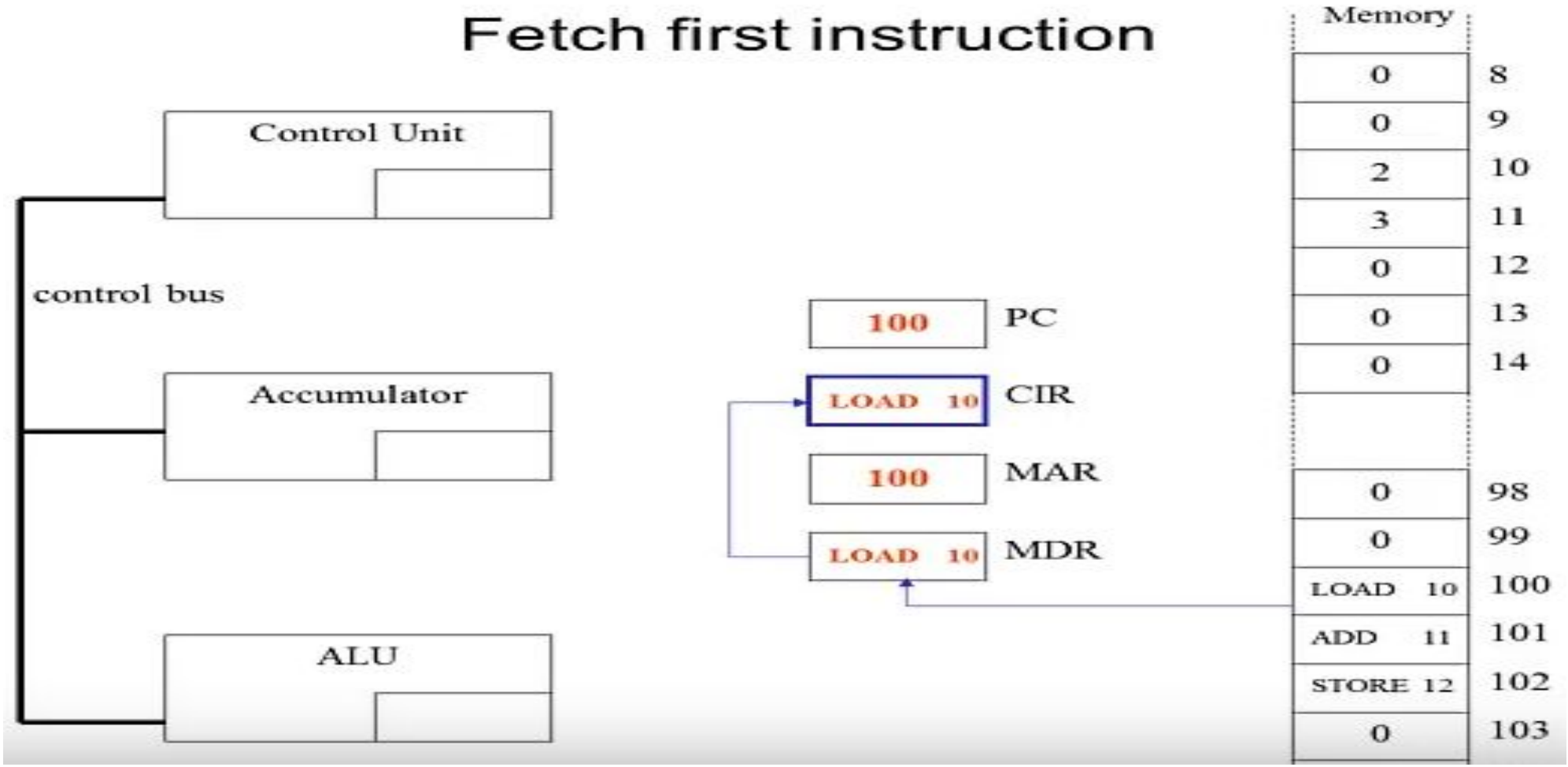

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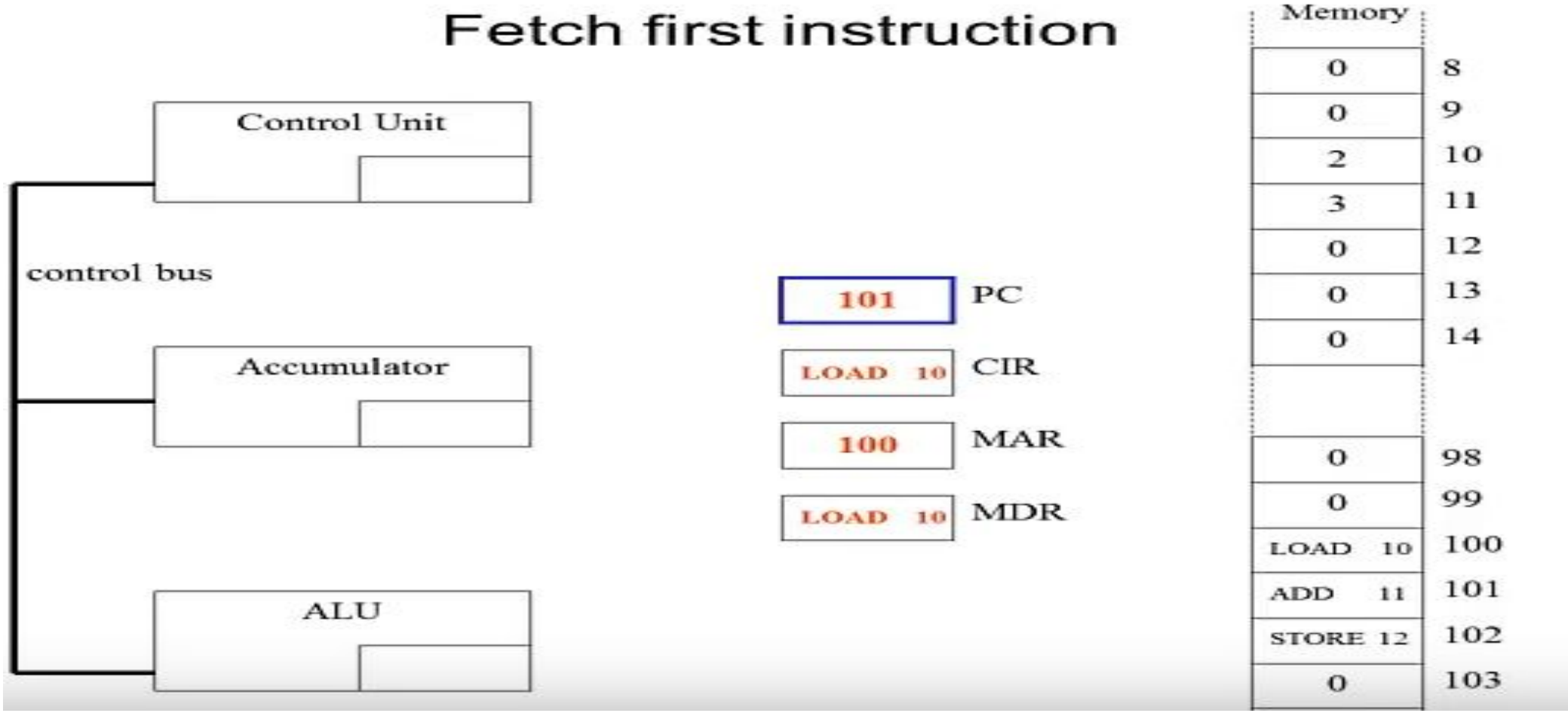

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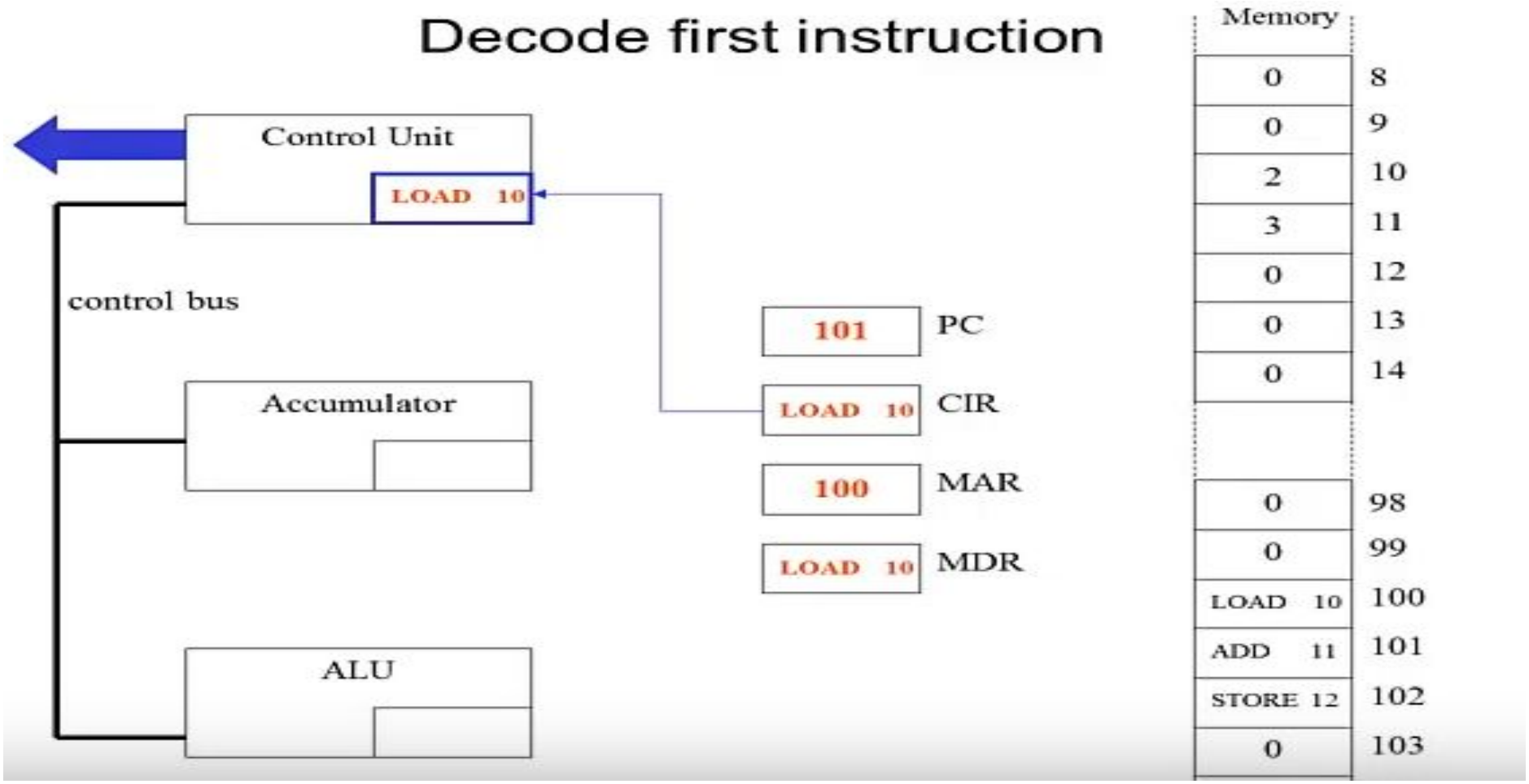

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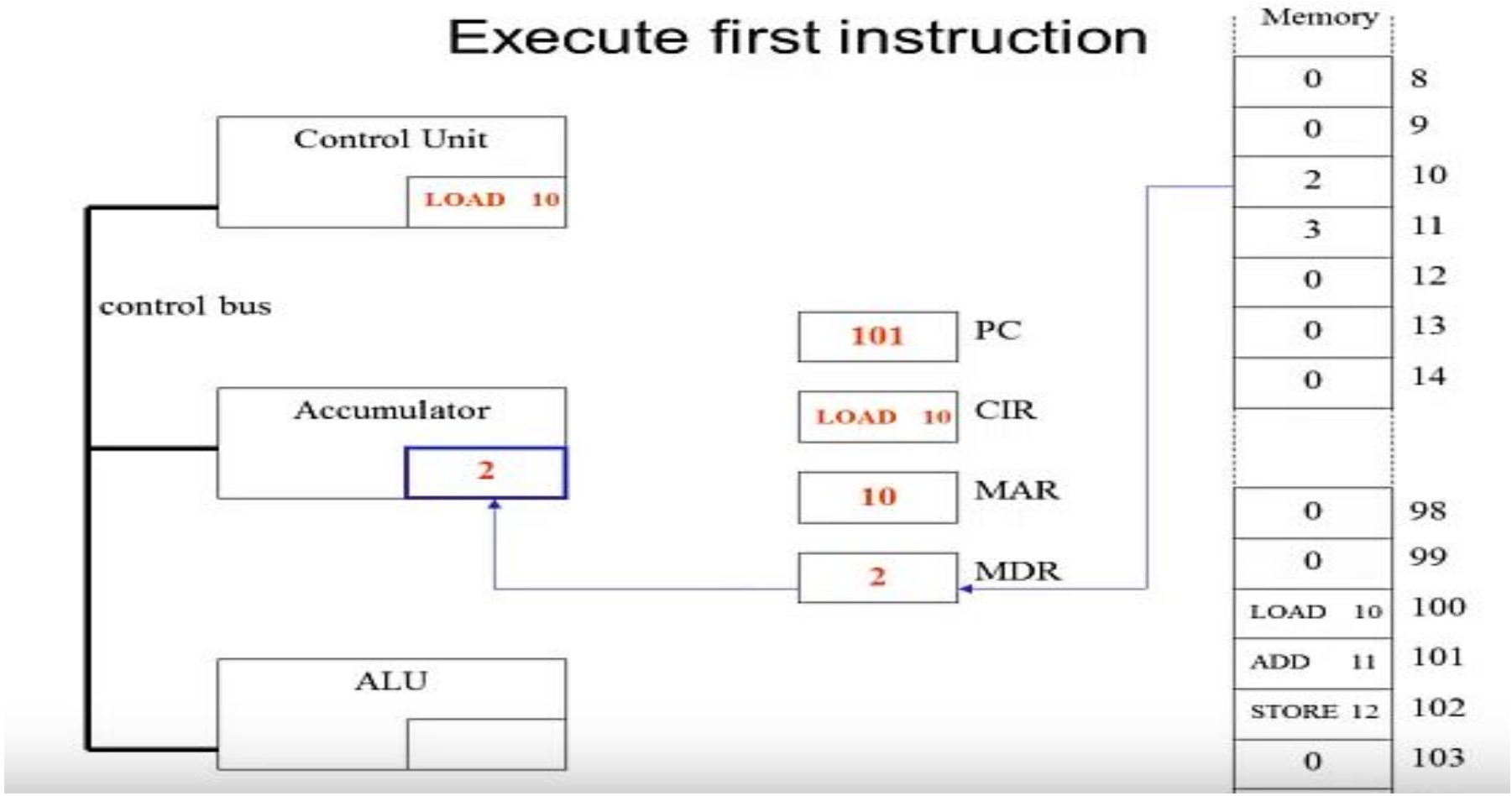
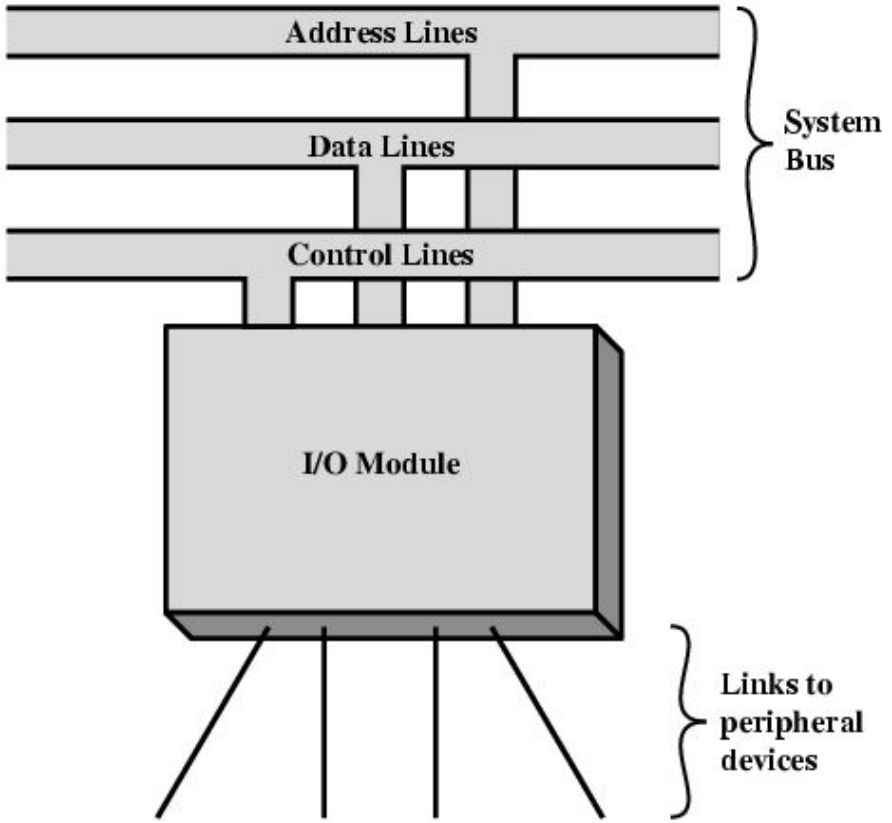


External Devices

- Broadly classify into three categories
  - Human Readable – VDTs and Printers.
  - Machine Readable – magnetic disk and tape system, sensors.
  - Communication – terminals, modem, NIC and other machines.
- Suitable for communicating with computer user – human readable.
- Suitable for communicating with equipment – machine readable.
- Suitable for communicating with remote devices – communication.

External Devices

- Provide a means of exchanging data.
- It connects through a link to an I/O module.
  - Used to exchange control, status, and data between the I/O module and external device.
- External device is referred to us peripheral device or a peripheral.



External Devices

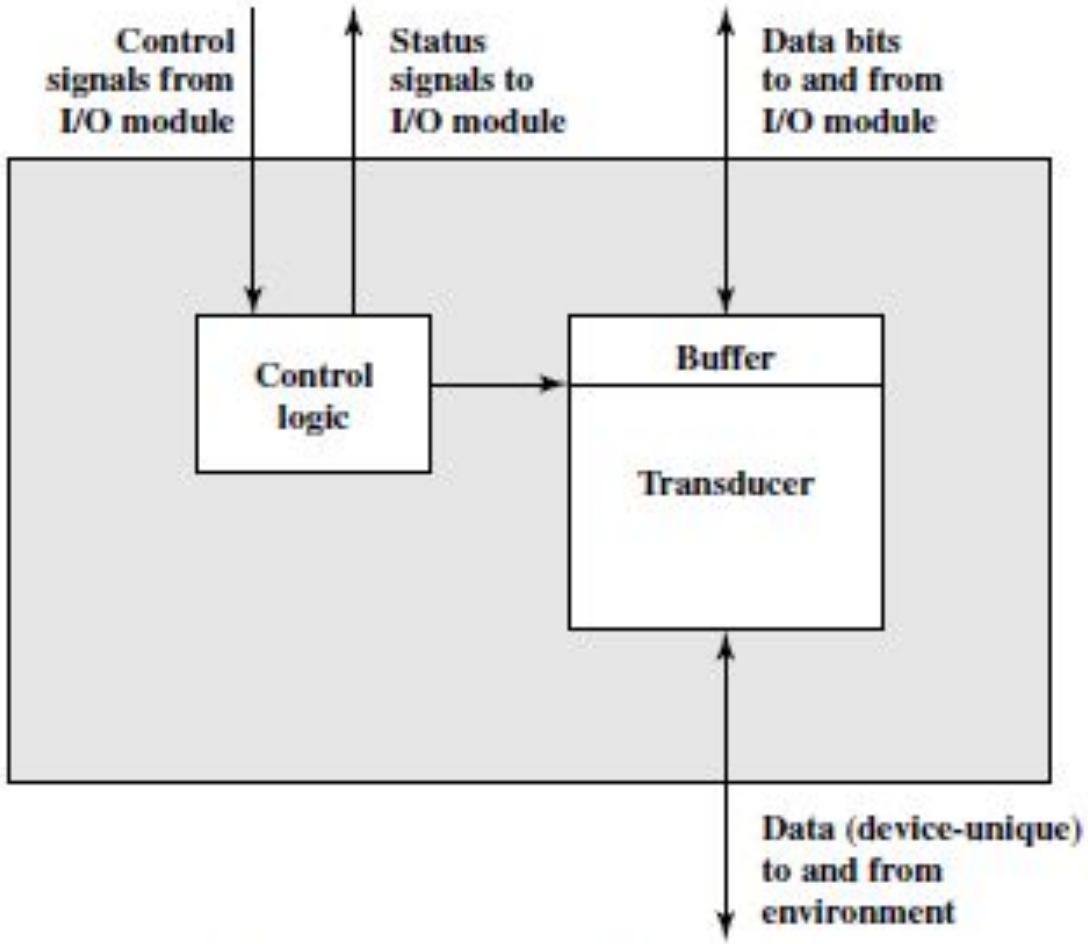


Figure 7.2 Block Diagram of an External Device

External Devices

- Interface to the I/O module is in the form of
  - Control signals, data, and status signals.
- Control signals determine the function that the device will perform.
  - Transfer data to the I/O module, accept data from I/O module, report status, or perform some control function particular to the device.
- Data are in the form of bits to be sent to or received from I/O module.
- Status signals indicate the state of the device.

I/O Modules

Module Functions

- Major functions or requirements for an I/O module
  - Control and Timing
  - Processor communication
  - Device communication
  - Data buffering
  - Error detection

I/O Modules

Control and Timing

- Coordinates the flow of traffic between internal resources and external devices.
- From external device to the processor might involve.
  - Processor questions the I/O module to check the status.
  - The I/O module returns the status.
  - If READY, processor requests the transfer of data, by mean of command to the I/O module.
  - I/O module obtains a unit of data from the external device.
  - The data is transferred from I/O module to the processor.

I/O Modules

Processor Communication

- Command decoding – I/O modules accepts commands from the processor, sent as a signals on the control bus.
- Data – exchanged between the processor and the I/O module over the data bus.
- Status reporting – status of peripheral is reported to processor.
- Address recognition – I/O module must recognize one unique address for each peripheral it controls.

I/O Modules

I/O Module Structure

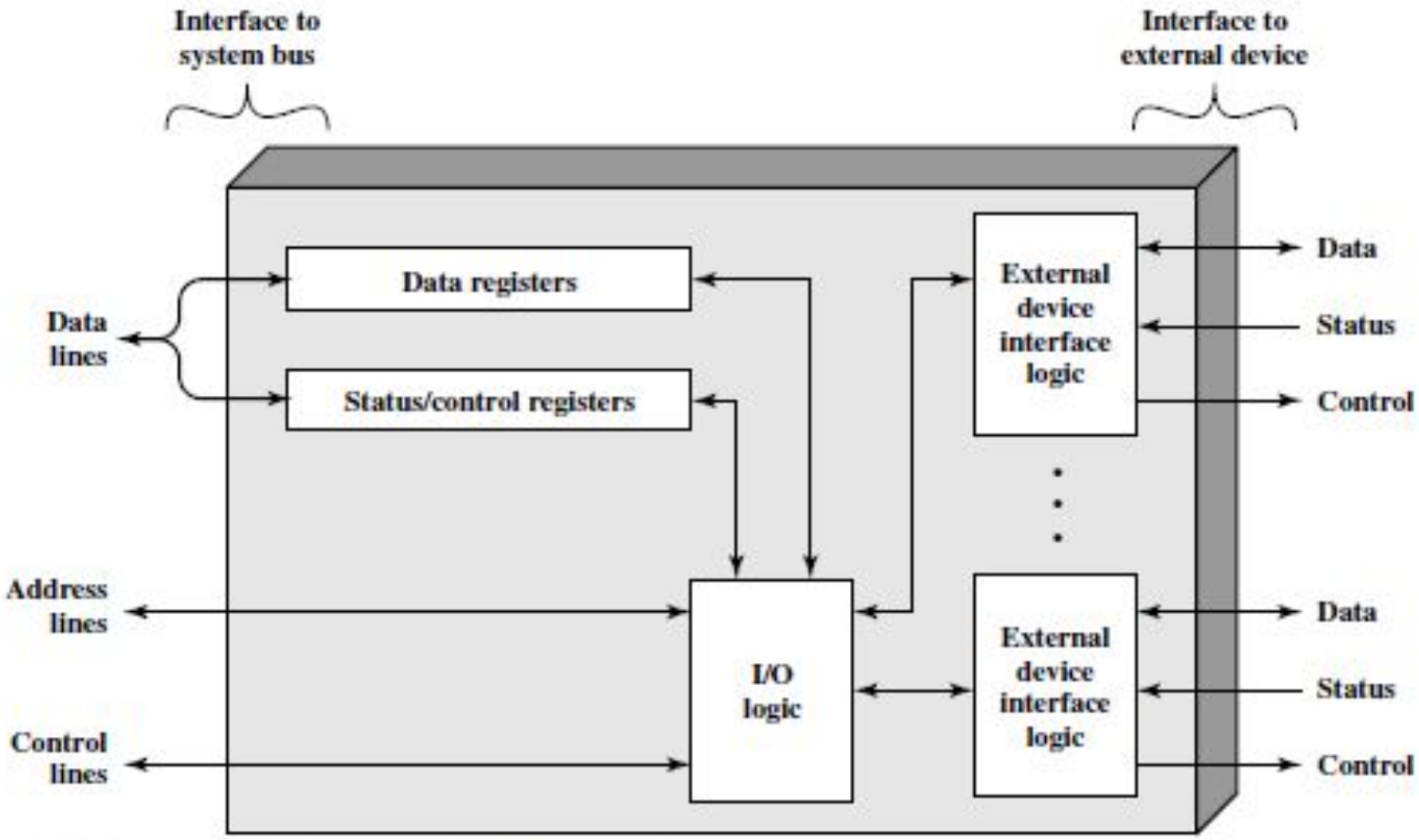


Figure 7.3 Block Diagram of an I/O Module

Programmed I/O

- Three possible techniques for I/O operations.

1. With programmed I/O, data are exchanged between the processor and I/O module.
2. processor executes program that provides direct control of I/O operations.
3. Processor must has to wait for issuing a command until I/O operation is complete.

Programmed I/O

- With interrupt-driven I/O, the processor issues an I/O command
  - Continues its other instructions and is interrupted by the I/O module when completed.
  - In both techniques the processor has to take data in and out from main memory.
- Alternative approach, direct memory access
  - Memory and I/O module directly exchanges the data without the intervention of processor.

Programmed I/O

Table 7.1 I/O Techniques

	No Interrupts	Use of Interrupts
I/O-to-memory transfer through processor	Programmed I/O	Interrupt-driven I/O
Direct I/O-to-memory transfer		Direct memory access (DMA)

Programmed I/O

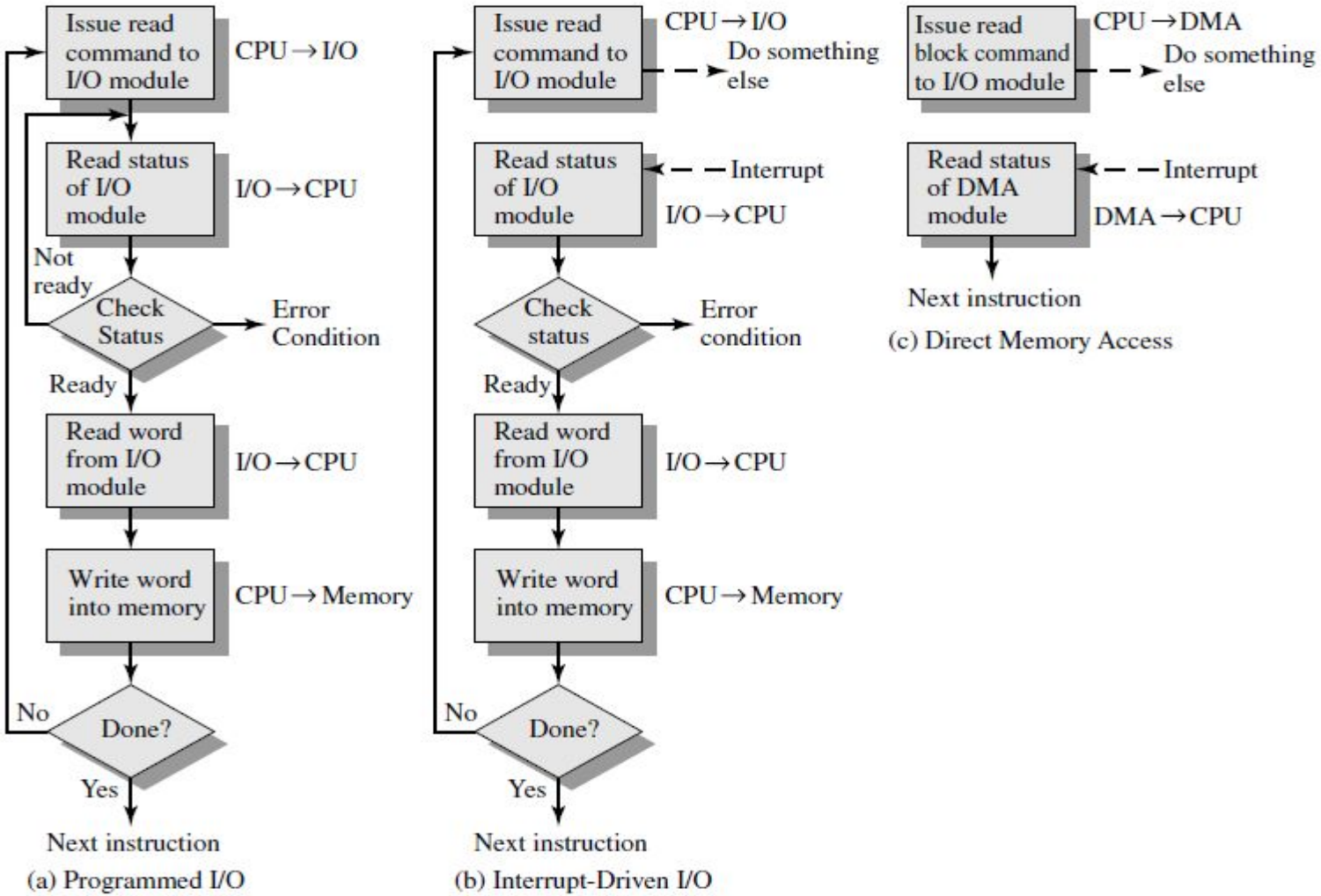


Figure 7.4 Three Techniques for Input of a Block of Data

Programmed I/O

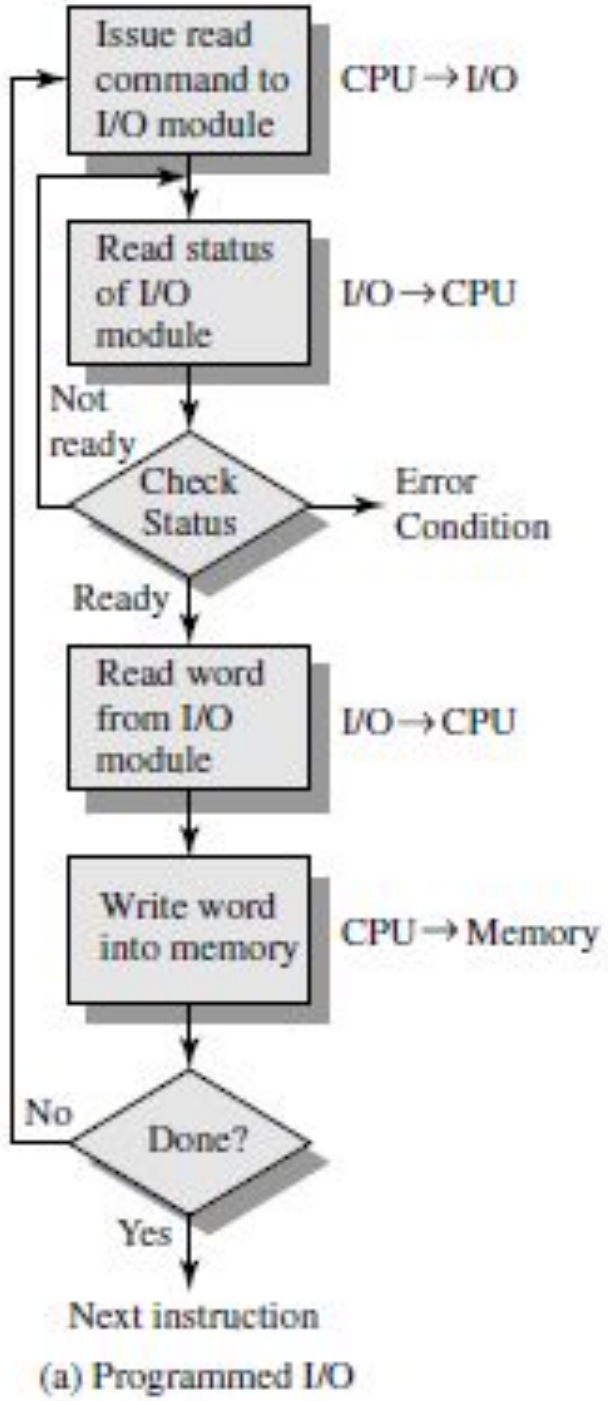
I/O Commands

- To execute I/O related instruction, processor issues
  - Address specifying the I/O module and external device with an I/O command.
- Four types of commands may receive by I/O module
  - Control
  - Test
  - Read
  - Write

Programmed I/O

I/O Commands

- Use of programmed I/O to read in a block of data from a peripheral device into memory.



(a) Programmed I/O

Programmed I/O

I/O Instructions

- Close correspondence between I/O instructions and I/O commands.
  - Instructions are easily mapped into I/O commands.
  - The form of instruction depends on the way in which external devices are addressed.
- Many I/O devices connected though I/O modules to the system.
  - Each device has a unique identifier.
  - Command contains the address of the device.
  - Each module interprets the address lines to determine his command.

Programmed I/O

An Example of a RISC-Style I/O Program

	Move	R2, #LOC	Initialize pointer register R2 to point to the address of the first location in main memory where the characters are to be stored.
	MoveByte	R3, #CR	Load ASCII code for Carriage Return into R3.
READ:	LoadByte	R4, KBD_STATUS	Wait for a character to be entered.
	And	R4, R4, #2	Check the KIN flag.
	Branch_if_[R4]=0	READ	
	LoadByte	R5, KBD_DATA	Read the character from KBD_DATA (this clears KIN to 0).
	StoreByte	R5, (R2)	Write the character into the main memory and
	Add	R2, R2, #1	increment the pointer to main memory.
ECHO:	LoadByte	R4, DISP_STATUS	Wait for the display to become ready.
	And	R4, R4, #4	Check the DOUT flag.
	Branch_if_[R4]=0	ECHO	
	StoreByte	R5, DISP_DATA	Move the character just read to the display buffer register (this clears DOUT to 0).
	Branch_if_[R5]≠[R3]	READ	Check if the character just read is the Carriage Return. If it is not, then branch back and read another character.

Programmed I/O

An Example of a CISC-Style I/O Program

	Move	R2, #LOC	Initialize pointer register R2 to point to the address of the first location in main memory where the characters are to be stored.
READ:	TestBit	KBD_STATUS, #1	Wait for a character to be entered
	Branch=0	READ	in the keyboard buffer KBD_DATA.
	MoveByte	(R2), KBD_DATA	Transfer the character from KBD_DATA into the main memory (this clears KIN to 0).
ECHO:	TestBit	DISP_STATUS, #2	Wait for the display to become ready.
	Branch=0	ECHO	
	MoveByte	DISP_DATA, (R2)	Move the character just read to the display buffer register (this clears DOUT to 0).
	CompareByte	(R2)+, #CR	Check if the character just read is CR (carriage return). If it is not CR, then
	Branch≠0	READ	branch back and read another character. Also, increment the pointer to store the next character.

Summery

- External Devices
- I/O Modules
- Programmed I/O
  - An Example of a RISC-Style I/O Program
  - An Example of a CISC-Style I/O Program

Thank You

For your Patience